

EART11200 - The Natural Scientist's Toolkit
Mathematics and Physics

Week 4 Lecture - Optimisation and Properties of Derivatives

Last Week...

- Use the **Chain Rule** to differentiate:

$$f(x) = \sin(3x + 1)$$

- **Step 1**: Split the function and substitute

$$f(g) = \sin(g) \qquad g(x) = 3x + 1$$

- **Step 2**: Differentiate:

$$\frac{df}{dg} = \cos(g) \qquad \frac{dg}{dx} = 3$$

- **Step 3**: Apply the product rule and substitute back in:

$$f(x) = 3\cos(3x + 1)$$

The Chain Rule

$$\frac{df}{dx} = \frac{df}{dg} \times \frac{dg}{dx}$$

Function $y = f(x)$	Derivative $\frac{dy}{dx}$
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

Last Week...

- Use the **Product Rule** to differentiate:

$$f(x) = x\exp(3x)$$

- **Step 1**: Split the function:

$$u(x) = x \qquad v(x) = \exp(3x)$$

- **Step 2**: Differentiate:

$$\frac{du}{dg} = 1 \qquad \frac{dv}{dx} = 3\exp(3x)$$

- **Step 3**: Apply the product rule and substitute back in:

$$f(x) = 3x\exp(3x) + \exp(3x) = (3x+1)\exp(3x)$$

The Product Rule

$$\frac{df}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

Function $y = f(x)$	Derivative $\frac{dy}{dx}$
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

Last Week...

- Use the **Quotient Rule** to differentiate:

$$f(x) = \frac{3x+1}{6x-1}$$

- **Step 1**: Split the function:

$$u(x) = 3x+1 \qquad v(x) = 6x-1$$

- **Step 2**: Differentiate:

$$\frac{du}{dg} = 3 \qquad \frac{dv}{dx} = 6$$

- **Step 3**: Apply the product rule and substitute back in:

$$f(x) = \frac{3(6x-1) - 6(3x+1)}{(6x-1)^2} = -\frac{9}{(6x-1)^2}$$

The Quotient Rule

$$\frac{df}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Function $y = f(x)$	Derivative $\frac{dy}{dx}$
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
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Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

Intended Learning Outcomes

(1) Describe the properties of derivatives in the analysis of functions.

(2) Analyse the behavior of functions by applying differentiation in theoretical and real-world applications.

(3) Solve differentiated functions to determine stationary values in theoretical and real-world applications.

(4) Apply differentiation techniques to optimise functions in real-world problems.

Properties of Derivatives

- For the last couple of weeks, we have been learning how to calculate derivatives of functions.
- Whilst telling us the rate of change of a function, derivatives can also be used to determine certain properties of a function!
- We now carry on with the topic of differentiation onto the topic of finding maxima and minima.
- This means to find the point on a curve that is either a maximum value or a minimum value. This turns out to be very useful.
- More generally maxima and minima are called stationary values or stationary values.

Properties of Derivatives

- Finding **maximum**, **minimum** or **saddle point** of a function, or optimising a function, requires a number of steps:
- We see that for a function $f(x)$:
  - If $f'(x) > 0$ then $f(x)$ is increasing.
  - If $f'(x) > 0$ then $f(x)$ is decreasing.
  - If $f'(x) = 0$ it is likely that $f(x)$ will be changing direction!
- Once we know where the these stationary values are for a function, further analysis can help classify them.

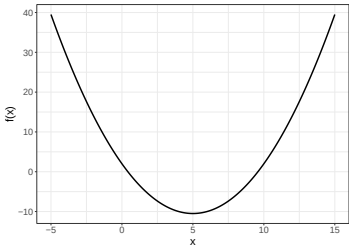
Example: Quadratic function

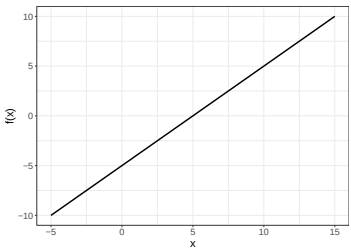
- Consider a quadratic function

$$f(x) = \frac{x^2}{2} - 5x + 2$$

- The thing to notice about the derivative from the graph, is that it is equal to zero at the point where $f(x)$ is at its minimum (or stationary).
- Differentiate to find the stationary value

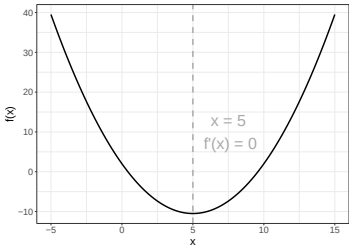
$$\frac{df}{dx} = x - 5$$

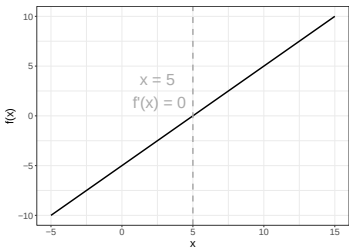




Example: Quadratic function

- Find the minimum by setting the derivative to zero and solve:
$$\frac{df}{dx} = 0 \implies x - 5 = 0 \implies x = 5$$
- This tells us that the minimum value does indeed occur at $x = 5$.
- We can evaluate the value of $f(x)$ at the stationary value (i.e. maximum and minimum)
$$f(5) = \frac{5^2}{2} - 5 \times 5 + 2 = 10.5$$





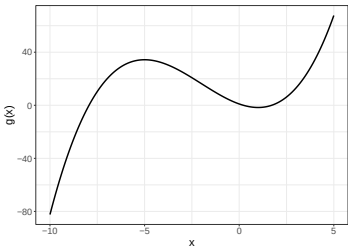
Example: Cubic function

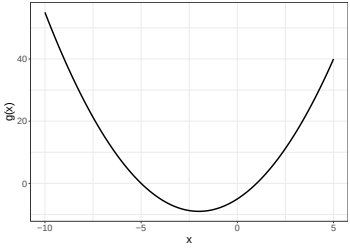
- Sometimes there can be more than one stationary value. Consider a cubic equation:

$$f(x) = \frac{x^3}{3} + 2x^2 - 5x + 1$$

- We can see that there is a point on the curve that where a minimum occurs ($x = 2$) and where a maximum occurs ($x = -5$), not immediately visible.
- Differentiate to find the stationary values:

$$\frac{df}{dx} = x^2 + 4x - 5$$



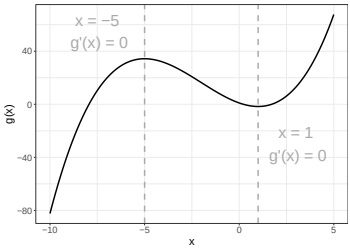


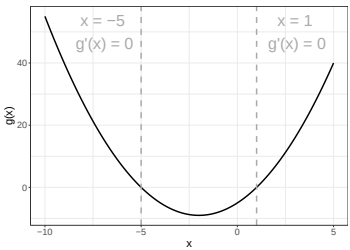
Example: Cubic function

- Find the minimum and maximum by setting the derivative to zero and solve

$$\frac{df}{dx} = 0 \implies x^2 + 4x - 5 = 0 \implies (x-1)(x+5) = 0$$

- This tells us that two stationary values at $x = -5$ and $x = 1$.
- It's more difficult when there are multiple stationary values to determine whether they are maxima, minima or saddle points, but for now, the plot confirms that $x = -5$ and $x = 1$ are a maximum and minimum respectively.



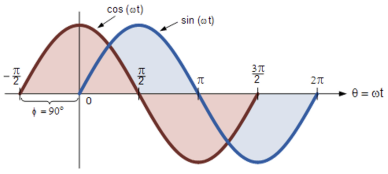


EXAMPLE: $\cos(x)$ AND $\sin(x)$

- We know that from the laws of calculus that

$$y = \sin(x) \implies \frac{dy}{dx} = \cos(x)$$

- We can see that the properties of derivatives follow from plotting the two functions against each other:
  - $\cos(x) = 0$ when $\sin(x)$ is stationary
  - $\cos(x) > 0$ when $\sin(x)$ is increasing
  - $\cos(x) < 0$ when $\sin(x)$ is decreasing



Properties of Derivatives

- In the two previous examples, we determined whether stationary values were maxima, minima, or saddle points from a plot. We could and strictly should verify this.
- Furthermore, both examples were simple enough that we could solve for zero just by rearranging.
- We'll come to both of these issues next week!
- An informal way of checking for a maximum or a minimum is by using the properties of the derivative.

Example: Quadratic function

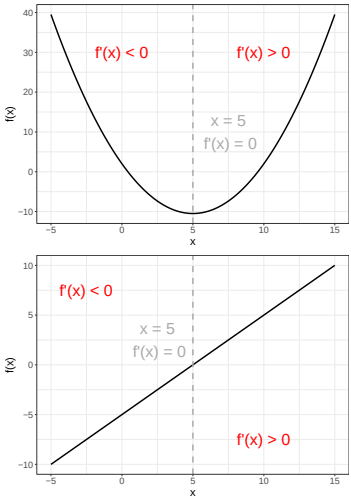
- Let's return to a quadratic function

$$f(x) = \frac{x^2}{2} - 5x + 2 \qquad \frac{df}{dx} = x - 5$$

- We had a stationary value at $x = 5$, so evaluate the derivative either side:

$$x = 4 \implies \frac{df}{dx}(4) = x - 5 = -1$$
$$x = 6 \implies \frac{df}{dx}(6) = 6 - 5 = 1$$

- Therefore $f(x)$ is decreasing when $x < 5$ and increasing when $x > 5$ so we do indeed have a minimum.



Optimisation

- Optimisation problems involve finding the best solution, often the maximum or minimum value(s) of a function given set of constraints.
- They are quite common in real-world applications:
  (a) a function describing gross profit and a function describing costs, then you can optimise net profit for maximum gain!
  (b) want to construct the largest box possible but have a limit on the size of a piece of cardboard, you can optimise the boxes volume!
- Derivatives and their properties play a crucial role in a wide range of problems such as optimisation.
- We study a function to find the maxima and minima which "optimises" it, which can play a vital role in decision making.
- Lets consider an example.

EXAMPLE: GOLD SEAM PROBLEM

- You decide to mine a gold seam and want to decide on how long to mine it for to maximise your profits.
- The value of gold you can mine per day (in millions), G, is a function of the number of days, n, that you mine for

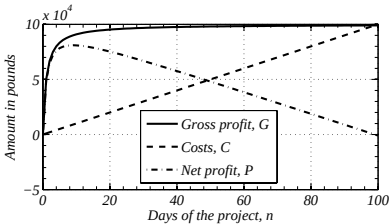
$$G(n) = 100 - \frac{100}{n+1}$$

- The cost of renting the equipment to mine, $C(n)$, increases with the number of days

$$C(n) = n$$

- Your profit is given by:

$$P(n) = G(n) - C(n) = 100 - \frac{100}{n+1} - n$$



EXAMPLE: GOLD SEAM PROBLEM

- Can we optimise the net profit? Yes, using differentiation and using its derivative to look for stationary values!
- Profit is given by

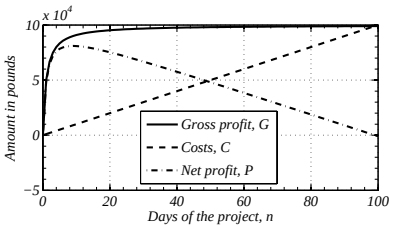
$$P(n) = 100 - \frac{100}{n+1} - n$$

- Find the derivative

$$\frac{dP}{dn} = 0 + \frac{d}{dn}\left(\frac{10}{n+1}\right) - 1$$

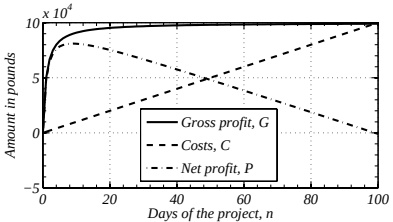
- Using the chain rule we see that

$$\frac{d}{dn}\left(\frac{10}{n+1}\right) = -\frac{10}{(n+1)^2}$$



Example: Gold seam problem

- So then
$$\frac{dP}{dn} = \frac{100}{(n+1)^2} - 1$$
- Where are the stationary values? Set the derivative to zero and solve!
$$\begin{aligned}\frac{dP}{dn} = 0 &\implies \frac{100}{(n+1)^2} - 1 = 0 \\ &\implies (n+1)^2 = 100 \\ &\implies n^2 + 2n - 99 = 0 \\ &\implies (n+11)(n-9) = 0\end{aligned}$$



Example: Gold seam problem

- The stationary values are at $n = -11$ and $n = 9$. We ignore the first as we cannot have negative time! We can see from the graph that at $n = 9$ $P(n)$ is maximised!
- Therefore, we optimise the profit at 9 days so we should stop digging after that.
- We have determined whether stationary values

